

OE-EC704A — Web Technology

Semester VII • 3L:0T:0P • 3 Credits

Module 2 Exam Notes: Introduction to Java Programming & OOP

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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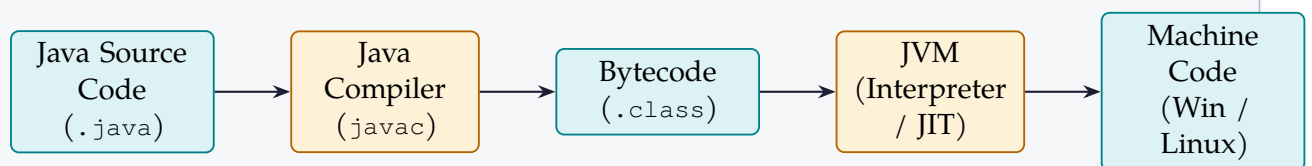
1. Java Execution Environment

Java uses a hybrid execution model combining compilation and interpretation. This ensures its core principle: **Write Once, Run Anywhere (WORA)**.

Bytecode and Platform Independence

- **Bytecode:** A highly optimized set of machine instructions designed for execution by the JVM. It is stored in a `.class` file after compilation.
- **Platform Independence:** Java source code compiles into platform-independent bytecode. However, the **JVM itself is platform-dependent** because it must translate bytecode into the host system's native machine instructions.

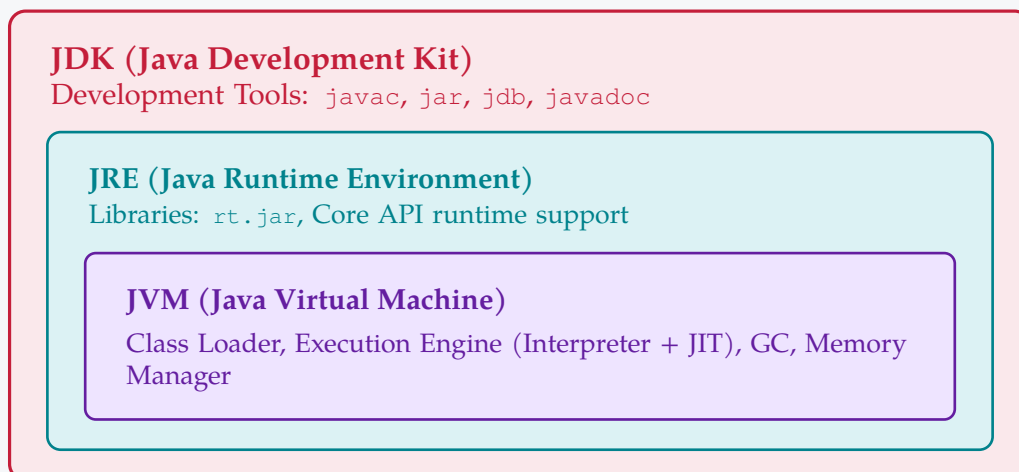
Java Compile-and-Run Execution Flow



1.1 JDK vs. JRE vs. JVM

Understanding the boundary between JVM, JRE, and JDK is crucial:

JDK, JRE, and JVM Containment Relationship



Metric	JVM (Virtual Machine)	JRE (Runtime)	JDK (Development)
Composition	Classloader, Execution Engine, Garbage Collector.	JVM + Core class libraries (<code>rt.jar</code>) + plugins.	JRE + Compiler (<code>javac</code>) + tools (<code>jar</code> , <code>jdb</code>).
Purpose	Execution of compiled bytecode.	Running Java applications.	Developing and running Java apps.
Target User	None (internal engine).	End users/clients.	Developers.

2. Java Characteristics and Features

Java's design is guided by a set of primary characteristics (often called "Java Buzzwords"):

Core Java Characteristics

1. **Simple:** Clean syntax derived from C++ but without complex features like pointers, operator overloading, or multiple inheritance.
2. **Object-Oriented:** Everything in Java is an object (except primitive types), promoting modularity and extensibility.
3. **Robust:** Emphasizes early error checking via compile-time checking, runtime exception handling, and automatic garbage collection.
4. **Secure:** Executes inside a virtual machine sandbox without direct memory manipulation or pointers, preventing buffer overflows and unauthorized system access.
5. **Portable and Architecture-Neutral:** Compiles to bytecode that can run on any device with a compatible JVM, without needing recompilation.
6. **Multithreaded:** Built-in support for concurrent execution of multiple tasks, improving resource utilization in GUI and server applications.

3. Data Types, Operators, and Expressions

3.1 Primitive vs. Reference Data Types

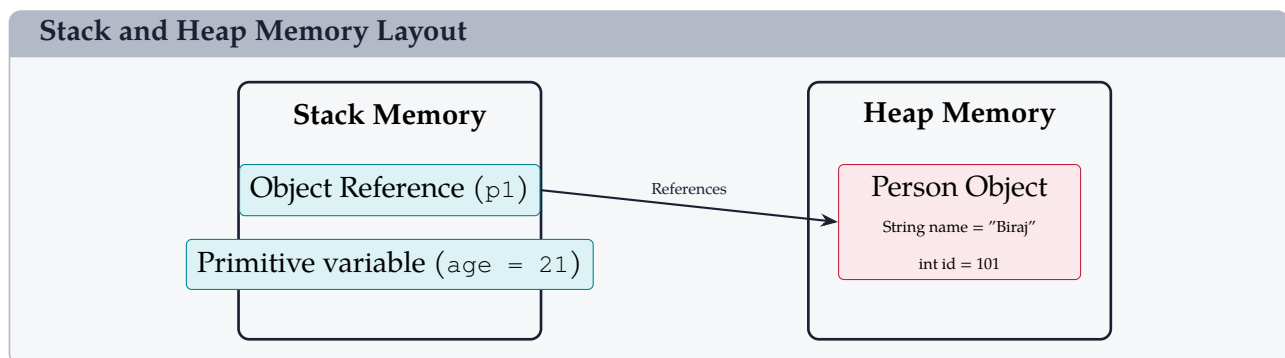
Java enforces static type checking and groups types into two primary categories:

Feature	Primitive Data Types	Reference Data Types
Allocation	Stored directly on the stack memory.	Object on Heap; address reference on Stack.
Default Value	0, 0.0, or false (never null).	null by default.
Size	Fixed sizing (e.g., int: 32-bit, double: 64-bit).	Depends on the host system architecture.
Examples	byte, short, int, long, float, double, char, boolean	Classes, interfaces, arrays, strings.

3.2 Memory Allocation Model

Java allocates memory across two distinct spaces during runtime:

- **Stack Memory:** Stores primitive variables and local references. Uses LIFO (Last In First Out) allocation.
- **Heap Memory:** Dynamic memory pool that hosts all runtime objects and instance variables.



3.3 Operators and Expressions

Java provides a comprehensive set of operators to build computational expressions:

- **Arithmetic Operators:** Binary (+, -, *, /, %) and Unary (++ , --).
- **Relational Operators:** Compare operands for equality or order (==, !=, <, >, <=, >=).
- **Logical Operators:** Combine boolean expressions (&& (logical AND), || (logical OR), ! (logical NOT)). Supports short-circuit evaluation.
- **Bitwise Operators:** Perform bit-level manipulations (&, |, ^, ~, <<, >>, >>>).
- **Ternary Operator:** Conditional assignment shorthand (variable = (condition) ? valueIfTrue : valueIfFalse).

4. Control Flow and Branching

Java supports selection, iteration, and jump statements to control execution paths:

- **Selection Statements:** `if-else` blocks for boolean conditions and `switch` blocks for multi-way branching (supports `byte`, `short`, `char`, `int`, `Enums`, and `String`).
- **Iteration Statements:**
 - `for` loops for index-based loops.
 - `while` loops for condition-controlled loops (pre-test).
 - `do-while` loops for exit-controlled loops (post-test, runs at least once).
- **Jump Statements:**
 - `break`: Terminated loop or switch block immediately.
 - `continue`: Skips the rest of the current loop body and proceeds to the next iteration step.
 - Labeled Statements: Loops can be labeled (e.g., `outerLoop: for(...)`) to specify which loop a nested `break` or `continue` targets.

Quick Revision — Module 2

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Java Basic Syntax & Loop Structures

Concept	Syntax and Example Usage
Simple Program	<pre>public class Test { public static void main(String[] args) { ... } }</pre>
Data Type Sizing	byte (8-bit), char (16-bit Unicode), int (32-bit), double (64-bit)
Ternary Operator	<pre>int min = (a < b) ? a : b;</pre>
Short-Circuit logic	<pre>if (obj != null && obj.isValid()) { ... } // prevents NPE</pre>
Loop Statements	Conditional and Iterative Formats
for Loop	<pre>for(int i = 0; i < n; i++) { ... }</pre>
while Loop	<pre>while(condition) { statements; }</pre>
do-while Loop	<pre>do { statements; } while(condition);</pre>
Labeled Break	<pre>outer: for(...) { if(cond) break outer; }</pre>

One-Line Definitions

- **JVM (Java Virtual Machine):** The abstract execution engine that interprets and executes compiled Java bytecode.
- **Bytecode:** Platform-independent intermediate code representation generated by `javac`.
- **JDK:** Java Development Kit containing compiler, tools, and the JRE runtime.
- **JRE:** Java Runtime Environment providing JVM and core system class libraries.
- **Stack Memory:** LIFO memory space storing local variables and function execution frames.
- **Heap Memory:** General dynamic memory pool where all Java objects are instantiated.
- **Short-Circuiting:** Logical expression evaluation that stops as soon as the outcome is guaranteed.
- **Platform Independence:** The ability of bytecode to run unmodified on any OS running a JVM.

Frequently Asked / Viva Questions

Q1. Why is Java platform-independent but JVM is platform-dependent?

Java bytecode is standardized and can run on any machine. However, the JVM must be custom-tailored to the target operating system to translate that bytecode to native CPU instructions.

Frequently Asked / Viva Questions (contd.)**Q2. What is JIT Compiler in Java?**

The Just-In-Time (JIT) compiler is a component of the JVM execution engine. It compiles frequently executed bytecode (hot spots) directly into native machine code at runtime to boost performance.

Q3. What is the difference between `path` and `classpath` variables?

`path` is an OS environment variable that locates shell commands like `javac` and `java`. `classpath` is a JVM-specific setting directing the classloader where to find compiled user class files.

Q4. What are the JVM memory partitions?

JVM memory is divided into: Method Area (class metadata, static variables), Heap (objects), Stack (local variables, execution frames), PC Register (current instruction address), and Native Method Stack.

Q5. Why does Java use Unicode for characters?

Java was designed for global applications. While ASCII uses 8 bits and represents 256 characters, Unicode uses 16 bits, allowing it to represent characters from all major languages.

Q6. What is the difference between prefix and postfix increment operators?

Prefix (`++i`) increments the variable first and then returns the incremented value. Postfix (`i++`) returns the original value first and then increments the variable.

Q7. Explain short-circuit logical operators.

In `A && B`, if `A` is false, the JVM skips evaluating `B` since the result is guaranteed to be false. Similarly, in `A || B`, if `A` is true, `B` is not evaluated.

Q8. Can we use a String in a switch-case statement?

Yes, starting from Java 7, String literals are supported in switch-case statements. The JVM compares strings using their `equals()` values and hash codes.

Q9. What is the difference between `break` and `continue` statements?

`break` terminates the loop completely, passing control to the statement after the loop. `continue` terminates only the current loop iteration, jumping to the loop update/condition.

Q10. Why is there no `goto` statement in Java?

Java does not support `goto` to enforce clean structured programming principles and prevent "spaghetti code." Instead, labeled `break` and `continue` statements are provided.

Java Iteration Statement Comparisons

Feature	for Loop	while Loop	do-while Loop
Loop Structure	Counter-based iteration.	Condition-controlled iteration.	Exit-controlled iteration.
Condition Check	Evaluated at start.	Evaluated at start.	Evaluated at end.
Min. Runs	0 runs if condition fails.	0 runs if condition fails.	Guaranteed at least 1 run.
Best Use Case	Iterating when count is known.	Loop until event triggers.	Interactive menus.